



12 TEMPERATURE & IDEAL GASES

H2 Physics 9478

12.1 Self-Check Questions

- S1** State what is meant by an "absolute temperature scale".
- S2** Explain the concept of absolute zero and its relation to the thermodynamic temperature scale.
- S3** What is "Avogadro's constant"?
- S4** State the ideal gas equation which uses
(i) number of moles,
(ii) number of molecules.
Explain the symbols used.
- S5** State the assumptions used for the derivation of the microscopic model of an ideal gas.
- S6** State the equation that relates pressure p , number density n , mass of a molecule m , and mean-square speed $\langle c^2 \rangle$.
- S7** State how the thermodynamic temperature of an ideal gas is related to the average microscopic kinetic energy of a molecule of the gas.

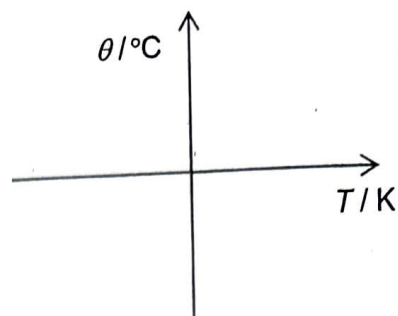
12.2 Self-Practice Questions

❖ Temperature Scales

- SP1** Express 40°C and 20°C as temperatures measured on the thermodynamic scale.
Hence, comment on the statement "Today's temperature is 40°C and yesterday was 20°C . So, it is twice as hot today as it was yesterday".

SP2 A student draws a linear graph on the axes shown in order to convert temperatures in Kelvins to temperatures in degrees Celsius.

What is the intercept on the vertical axis and the gradient of the line?



	vertical-intercept	gradient
A	0	$1/273.15$
B	-273.15	1
C	$+273.15$	1
D	0	273.15

SP3 Determine the approximate number of atoms in one cubic metre of ideal gas at a temperature of 27°C and a pressure of $1.0 \times 10^5 \text{ Pa}$.

❖ Equation of State

SP4 An ideal gas at pressure P (in pascal) and temperature T (in kelvin) has X molecules per unit volume.

If the pressure of the gas is reduced to $0.5P$ and its temperature is increased to $2T$, what is the number of molecules per unit volume of the gas in terms of X ?

❖ Kinetic Theory of Gases

SP5	$x\text{-velocity} / \text{m s}^{-1}$	-8.0	5.5	2.4	-3.1	6.0
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The x -velocities of several particles are given above.

Calculate the average x -velocity, the mean-square speed and the root-mean-square speed in the x -direction.

12.3 Discussion Questions

❖ Temperature Scales

- D1** A solid iron cube of side 20 mm has a mass of 6.3×10^{-3} kg. One mole of the iron atom has a mass of 5.6×10^{-2} kg.
- (a) How many iron atoms are there in the cube?
 - (b) Hence, find the maximum volume that an iron atom could occupy in the solid state.
 - (c) Use your answer in (b) to estimate the diameter of an iron atom and state any assumptions made.

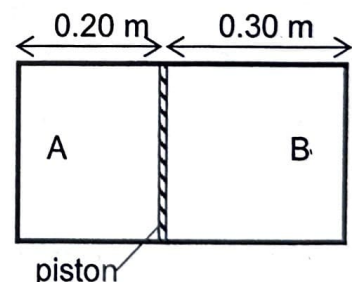
❖ Equation of State

- D2** An oven has a volume of 0.064 m^3 . The pressure and temperature of the air in the oven is $1.0 \times 10^5 \text{ Pa}$ and 27°C respectively. The mass of one mole of air is 0.030 kg. The air behaves as an ideal gas.
- (a) Calculate the mass of air in the oven.
 - (b) The oven is heated to a temperature of 180°C . The oven door is opened. Calculate the mass of air that must escape from the oven for the pressure in the oven to return to $1.0 \times 10^5 \text{ Pa}$.
- D3** A closed cylinder of length 0.50 m and cross-sectional area S is fitted with a smooth thermally insulating piston. An ideal gas is contained in each section A and B as shown.

Initially, the piston is in equilibrium at a distance of 0.20 m from one end of the cylinder, and the gas in each compartment is at a temperature of 27°C .

Now compartment A is submerged in a heat bath of 177°C , while the gas in B is maintained at 27°C . The piston moves through a displacement of x until a new position of equilibrium is obtained.

Determine x .



❖ Kinetic Theory of Gases

- D4 (a) A spherical balloon of volume 4000 cm^3 contains helium at a pressure of $1.20 \times 10^5 \text{ Pa}$. How many moles of helium are in the balloon, if the average kinetic energy of the helium atoms is $3.60 \times 10^{-22} \text{ J}$?
- (b) Determine the ratio of the total translational kinetic energy of the molecules in 1.0 mol of an ideal gas at room temperature (25°C) to the average kinetic energy of a sprinter, of mass 70 kg while running 100 m in 10 s .
- [N10/2/5; N85/2/3]
- D5 For a fixed amount of an ideal gas in a container with fixed walls, explain why pressure is proportional to mean-square speed of the molecules of the ideal gas.
- D6 (a) The temperature T of an ideal gas at pressure p is defined by the equation $p = nkT$. Identify the quantities n and k in this equation.
- (b) Another equation that relates pressure and mean-square speed is $p = \frac{1}{3} \rho_m \langle c^2 \rangle$. Identify the quantity ρ_m .
- D7 A certain plasma contains hydrogen ions (protons) and electrons in thermal equilibrium. Both protons and electrons can be assumed to behave like the particles of an ideal gas. The root-mean-square speed c_{rms} of the electrons in the plasma is estimated to be $3.0 \times 10^6 \text{ m s}^{-1}$.
- (a) Determine the root-mean-square speed c_{rms} of the hydrogen ions, given that the mass of a proton is $1.67 \times 10^{-27} \text{ kg}$ and that of an electron is $9.11 \times 10^{-31} \text{ kg}$.
- (b) Estimate the temperature of the plasma.
- D8 One mole of argon has a mass of 40 g . Assuming argon behaves as an ideal gas, calculate
- (a) the volume occupied by one mole of argon gas at a temperature of 273 K and a pressure of $1.01 \times 10^5 \text{ Pa}$,
- (b) the density of argon gas at this temperature and pressure,
- (c) the root-mean-square speed of its atoms at 273 K ,
- (d) the average kinetic energy of an argon atom.

[N99/III/4 part modified]

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- D9** The following passage is a quotation from the *Cambridge Encyclopaedia of Earth Science*.
In the upper layers of atmosphere, a large fraction of the hydrogen molecules travel so fast that they are able to escape completely from the Earth. It is estimated that about two-thirds of all the hydrogen molecules present in the atmosphere will escape in about one thousand years. There may also be some loss of helium, but for all other heavier molecules escape is negligible.

Suggest an explanation for each of the following observations.

- (a) Fast molecules are able to escape the atmosphere.
- (b) Molecules of small mass are able to escape whereas more massive ones are not.
- (c) Some molecules escape even though the speed required for escape is very much greater than their root-mean-square speed.
- (d) Large quantities of hydrogen have remained indefinitely on the Earth as a constituent of water.

[J89/2/10 part]

12.4 Challenging Questions

- C1** A tiny hole of area $8.0 \times 10^{-17} \text{ m}^2$ develops in a sealed glass tube of volume $2.0 \times 10^{-3} \text{ m}^3$. The original pressure in the tube is $1.0 \times 10^{-6} \text{ Pa}$. The air outside the tube has a pressure of $1.0 \times 10^5 \text{ Pa}$ and a temperature of 300 K.

(For this question, the mass of an air molecule may be taken as $5.6 \times 10^{-26} \text{ kg}$.)

- (a) Given that the rate S at which gas molecules hit an area A of the wall of their container is $S = \frac{1}{4} n c A$, where n is the number density of the gas molecules and c is their root-mean-square speed, calculate the rate at which air molecules leak into the tube from outside.
- (b) Determine how long will it take for the pressure in the tube to reach $5.0 \times 10^{-3} \text{ Pa}$.

[N00/2/3 part]

12.5 Numerical Answers

- D1** (a) 6.8×10^{22} , (b) $1.2 \times 10^{-28} \text{ m}^3$, (c) $4.9 \times 10^{-10} \text{ m}$
- D2** (a) 0.077 kg, (b) 0.026 kg
- D3** 0.05 m to the right
- D4** (a) 3.32 mol, (b) 1.1
- D7** (a) $7.0 \times 10^4 \text{ m s}^{-1}$, (b) $2.0 \times 10^5 \text{ K}$
- D8** (a) $2.25 \times 10^{-2} \text{ m}^3$, (b) 1.78 kg m^3 , (c) 413 ms^{-1} , (d) $5.65 \times 10^{-21} \text{ J}$
- C1** (a) $2.28 \times 10^{11} \text{ s}^{-1}$, (b) 2.95 h